

Python Developer Last-Minute Revision Notes

1. Python

Interpreted, high-level, dynamically typed language. Indentation matters.

Syntax: `print('Hello')`

Interview: Python is interpreted, object-oriented, cross-platform.

2. Variables & Data Types

`x=10; y=3.14; s='Hi'; flag=True`

Types: `int, float, str, bool, list, tuple, set, dict, None`

3. Input/Output

`name=input('Name: '); print(name)`

4. Operators

Arithmetic: `+-*/%//**` Comparison: `== != > < >= <=` Logical: `and or not` Membership: `in`

Identity: `is` Bitwise: `& | ^ << >>`

5. if/elif/else

`if x>0: print('Positive')`

6. Loops

`for i in range(5): ... ; while cond: ... ; break continue pass`

7. Functions

`def add(a,b): return a+b ; lambda x:x*x`

8. Strings

Methods: `upper lower strip split replace find count startswith endswith`

9. Lists

Mutable. Methods: `append extend insert remove pop sort reverse`

10. Tuples

Immutable. Packing: `t=1,2` Unpacking: `a,b=t`

11. Dictionaries

`{'a':1}`; methods: `keys values items get update pop`

12. Sets

Unique values. add remove union intersection difference symmetric_difference

13. Type Casting

`int('5') float(5) str(5) list('abc')`

14. Exceptions

`try: ... except Exception: ... finally: ... raise ValueError()`

15. Files

`open('a.txt','r'/'w'/'a'); with open(...) as f:`

16. Modules

`import math; from math import sqrt`

17. Built-ins

`len range type sum max min sorted zip enumerate map filter`

Top 30 Quick Interview/Aptitude Questions

1. What is Python? — Know the definition, syntax, one example, and one use case.
2. Difference between list and tuple? — Know the definition, syntax, one example, and one use case.
3. List vs set? — Know the definition, syntax, one example, and one use case.
4. Dict key rules? — Know the definition, syntax, one example, and one use case.
5. What is None? — Know the definition, syntax, one example, and one use case.
6. What is lambda? — Know the definition, syntax, one example, and one use case.
7. Difference between `==` and `is`? — Know the definition, syntax, one example, and one use case.
8. What is slicing? — Know the definition, syntax, one example, and one use case.
9. Mutable vs immutable? — Know the definition, syntax, one example, and one use case.
10. What is `range()`? — Know the definition, syntax, one example, and one use case.
11. `append` vs `extend`? — Know the definition, syntax, one example, and one use case.
12. `pop` vs `remove`? — Know the definition, syntax, one example, and one use case.
13. `sort` vs `sorted`? — Know the definition, syntax, one example, and one use case.

14. What is list comprehension? — Know the definition, syntax, one example, and one use case.
15. What is *args? — Know the definition, syntax, one example, and one use case.
16. **kwargs? — Know the definition, syntax, one example, and one use case.
17. Purpose of finally? — Know the definition, syntax, one example, and one use case.
18. What is a module? — Know the definition, syntax, one example, and one use case.
19. What is a package? — Know the definition, syntax, one example, and one use case.
20. How to read a file? — Know the definition, syntax, one example, and one use case.
21. What is enumerate()? — Know the definition, syntax, one example, and one use case.
22. zip()? — Know the definition, syntax, one example, and one use case.
23. map()? — Know the definition, syntax, one example, and one use case.
24. filter()? — Know the definition, syntax, one example, and one use case.
25. Difference between // and /? — Know the definition, syntax, one example, and one use case.
26. What is pass? — Know the definition, syntax, one example, and one use case.
27. break vs continue? — Know the definition, syntax, one example, and one use case.
28. What are exceptions? — Know the definition, syntax, one example, and one use case.
29. What is indentation? — Know the definition, syntax, one example, and one use case.
30. Why use with open()? — Know the definition, syntax, one example, and one use case.

One-Page Cheat Sheet

Variables: `x=10`

`if/elif/else`

`for i in range(n):`

`while cond:`

`def f(a): return a`

`lambda x:x*x`

List:[] Tuple:() Dict:{k:v} Set:{1,2}

Common methods: `append()`, `pop()`, `keys()`, `values()`, `items()`, `add()`, `union()`, `split()`, `replace()`

File: `with open('f.txt','r') as f:`

Imports: `import math`

Built-ins: len range type sum max min sorted zip enumerate map filter